ARYAN MEENA

Data Quality and Model Validation Analyst | Python, SQL, Excel
araj7042@gmail.com | (857) 425-6779 | LinkedIn | GitHub | Portfolio | Boston, MA (open to relocation)

EDUCATION

Master of Science in Applied Data Analytics - Boston University | GPA 3.75/4.0 **May 2026**
Bachelor of Technology in Electronics and Communication Engineering - Indian Institute of Space Science and Technology **May 2023**
Relevant Coursework: Data Mining, Data Analytics and Visualization, Big Data Analytics, Foundations of Machine Learning
Honors: First runner-up of 40+ teams, BU MedAI Hackathon (team lead); 3D PET model at validation MAE 7.31 Centiloid (r 0.968)

PROFESSIONAL EXPERIENCE

Research Assistant, Quantitative Finance, Boston University | *Boston, MA* **Sep 2025 – May 2026**
Skills – Python (pandas, NumPy, SciPy), SQL, Excel, bootstrap testing, Git

- Traced a 15-year Dollar Index trading signal to bad opening prices in a free data feed by integrating and reconciling exchange, vendor and Federal Reserve data in Python. **Published as first author in Risks (MDPI).**
- Screened 37 financial price feeds with automated data validation checks and **flagged 10 faulty series** before any analysis.
- **Caught 11 model validation failures** in real backtests, including look-ahead bias and mis-annualized Sharpe ratios, each flagged by a dedicated automated test.
- Coded White's Reality Check, Hansen's SPA and Romano-Wolf tests from scratch to correct for 216 strategy variants.
- Rewrote the NumPy backtest engine behind 1,312 strategy configurations to run 17x faster with identical results.
- Wrote the SQL and Python behind a live research dashboard on a 12-table relational database (star schema, 7 views, 484K monthly rows) so non-specialists can check every result without code.

Graduate Teaching Assistant, Boston University | *Boston, MA* **Jan 2026 – May 2026**
Skills – Python, scikit-learn, PyTorch, statistics

- Held weekly office hours for 60+ graduate students training as data scientists in Advanced Machine Learning and Data Science with Python, debugging student code and explaining model results in plain language.

Machine Learning Research Intern, ISRO Space Applications Centre | *Ahmedabad, India* **Jun 2022 – Jul 2022**
Skills – PyTorch, U-Net, DeepLabV3+, QGIS

- Trained U-Net and DeepLabV3+ deep learning models in PyTorch to segment roads in 6,723 aerial images, and traced a drop in accuracy on Indian satellite imagery to a sensor resolution and metadata mismatch.
- Evaluated models on IoU instead of pixel accuracy, since class imbalance made accuracy misleading, and showed the optimizer choice (Adam vs RMSProp) moved results as much as the architecture.

PROJECTS

Multi-Asset Risk Engine | *Python, NumPy, SciPy* **GitHub**

- Backtested 8 Value at Risk (VaR) models over 2,796 days with Kupiec and Christoffersen tests. Only DCC-GARCH with an empirical tail passed conditional coverage, holding 13% less VaR than historical simulation.
- Fitted GARCH and DCC-GARCH models by maximum likelihood in SciPy, and showed a normal model understated 99% expected shortfall by about a third (11.1% vs 7.4%).

Service Operations Data Pipeline | *Python, SQL, AWS, PySpark* **Live site**

- Automated daily AWS reporting over 130K+ records that publishes only after **17 data quality checks** pass, with alarms for broken or stale data (S3, Glue Spark jobs, Athena, Step Functions).
- Found that two alarming team metrics were data artifacts (212 of 215 tickets still open), which stopped a wrong conclusion, and shipped the pipeline under Git version control with 81 automated tests in CI/CD.

Hospital Readmission Risk Model | *Python, LightGBM, Tableau* **Live dashboard**

- Validated a classification model on 101,766 patient encounters with patient-grouped cross-validation and beat a published 30-day readmission benchmark (AUC 0.689 vs 0.664).
- Audited calibration and fairness by age, race and sex, recommended age-specific thresholds after AUC fell from 0.733 to 0.631 across age bands, and wrote TRIPOD+AI model documentation.

PUBLICATIONS

Meena and Pinsky. Two Blind Spots in Backtest Validation. *Risks (MDPI), 14(11), 247, 2026.*
Meena et al. WikiFlow: Predicting Wikipedia Editor Dropout at Scale Using Distributed Machine Learning on Google Cloud. *Springer, 2026.*

TECHNICAL SKILLS

Programming: Python (pandas, NumPy, SciPy), SQL (joins, CTEs, window functions), PostgreSQL, R, Excel (pivot tables, XLOOKUP)
Data Quality and Risk Management: data validation, reconciliation, quality control, VaR, expected shortfall, GARCH, regression, bootstrap
Cloud and Tools: AWS (S3, Glue, Athena), Spark (PySpark), Bloomberg data, Excel Power Query, Power BI, Tableau, Git, Docker, CI/CD